

Homework 1

EE 474 — Introduction to Control Systems

Patrick Gordon • September 9, 2026

Due Wednesday, September 16, 2026

Problem 1.1b — Drebbel incubator — feedback loop

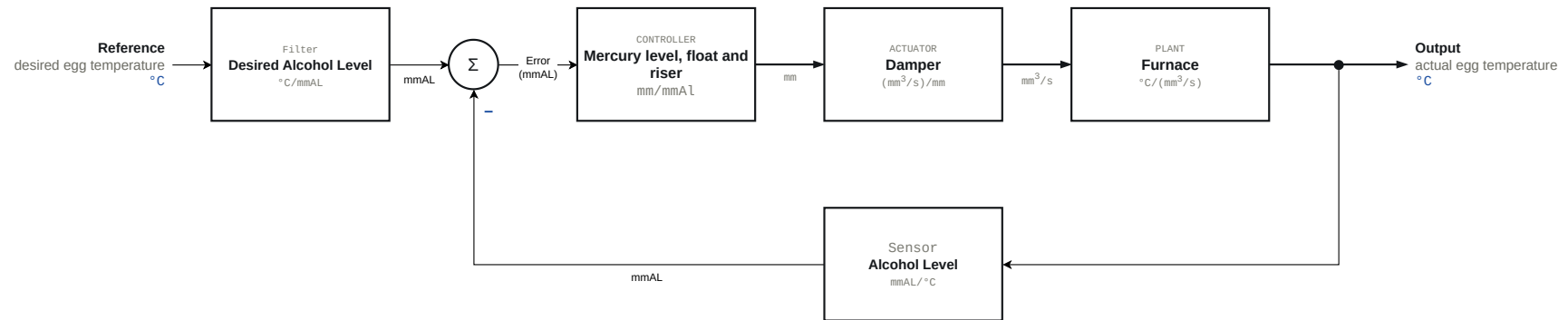


Figure 1: Block diagram of Drebbel's incubator drawn as a closed-loop control system.

Problem 1.1d — Watt's Engine — feedback loop

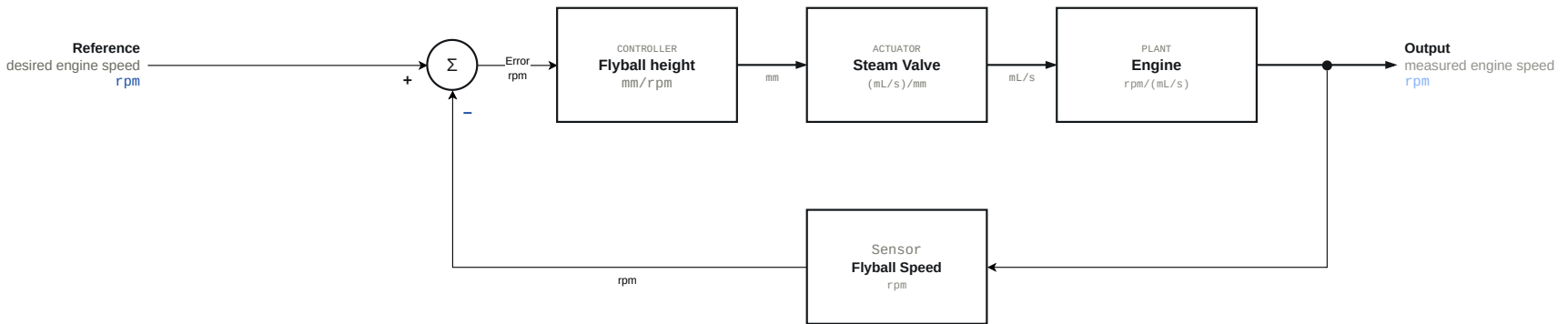


Figure 2: Block diagram of Watt's steam engine drawn as a closed-loop control system.

Problem 2.A — Vector operations

Code: hw_01_A.m

```
1 % Initial Vectors
2 a = [ 1 2 4 2 1];
3 b = [4 3 -1 0 1];
4
5 disp("A.1")
6 a+b
7
8 disp("A.2")
9 a - b
10
11 disp("A.3")
12 min(a)
13
14 disp("A.4")
15 (a>b)
16
17 disp("A.5")
18 (a==b)
19
20 disp("A.6")
21 sin(a)
22
23 disp("A.7")
24 exp(b)
```

Output: output/hw_01_A.txt

```
A.1
ans =

    5    5    3    2    2

A.2
ans =

   -3   -1    5    2    0

A.3
ans = 1

A.4
ans =

    0    0    1    1    0

A.5
ans =

    0    0    0    0    1

A.6
ans =

    0.8415    0.9093   -0.7568    0.9093    0.8415
```

A.7

ans =

54.5982	20.0855	0.3679	1.0000	2.7183
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Problem 2.B — Matrix operations

Code: hw_01_B.m

```
1 a = [1 2 3; 3 2 3; 0 1 0];
2 b = [ 0 0 1; 0 1 1; 1 1 0];
3 c = [1 2 3];
4
5 disp("B.1")
6 a + b
7
8 disp("B.2")
9 d = inv(b)
10
11 disp("B.3")
12 b * d
13
14 disp("B.4")
15 sin(a)
16
17 disp("B.5")
18 a * b
19
20 disp("B.6")
21 a .* b
22
23 disp("B.7")
24 a'
25
26 disp("B.8")
27 a + a'
```

Output: output/hw_01_B.txt

```
B.1
ans =

     1     2     4
     3     3     4
     1     2     0

B.2
d =

     1    -1     1
    -1     1     0
     1     0     0

B.3
ans =

     1     0     0
     0     1     0
     0     0     1

B.4
ans =
```

0.8415	0.9093	0.1411
0.1411	0.9093	0.1411
0	0.8415	0

B.5

ans =

3	5	3
3	5	5
0	1	1

B.6

ans =

0	0	3
0	2	3
0	1	0

B.7

ans =

1	3	0
2	2	1
3	3	0

B.8

ans =

2	5	3
5	4	4
3	4	0

Problem 2.C — Property of matrix times inverse

$b * d$ yields the identity function.

Problem 2.D — Matrix Multiplication

$a*b$ is a matrix product, where $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$

$a.*b$ is an element-wise multiplication where $c_{ij} = a_{ij} * b_{ij}$

Problem 2.E.1 — Roots of vector

Code: hw_01_E.m

```
1 a = [2 3 1];  
2  
3 disp("E.1")  
4 roots(a)
```

Output: output/hw_01_E.txt

```
E.1  
ans =  
  
-1.0000  
-0.5000
```

Problem 2.E.2 — Polynomial representation

$$2x^2 + 3x + 1$$

Problem 2.E.3 — Polynomial verification

Code: hw_01_E3.m

```
1 root = [-1; -0.5];  
2  
3 root_1_verification = 2*root(1)^2 + 3*root(1) + 1  
4 root_2_verification = 2*root(2)^2 + 3*root(2) + 1
```

Output: output/hw_01_E3.txt

```
root_1_verification = 0  
root_2_verification = 0
```

Problem 2.F — Simple Sinusoid Graph

Code: hw_01_F.m

```
1 t = linspace(0, 4, 1000);  
2 y = sin(2*pi*t);  
3  
4 figure;  
5 plot(t,y);  
6 xlabel('t');  
7 ylabel('y');  
8 title('Simple Sine');
```

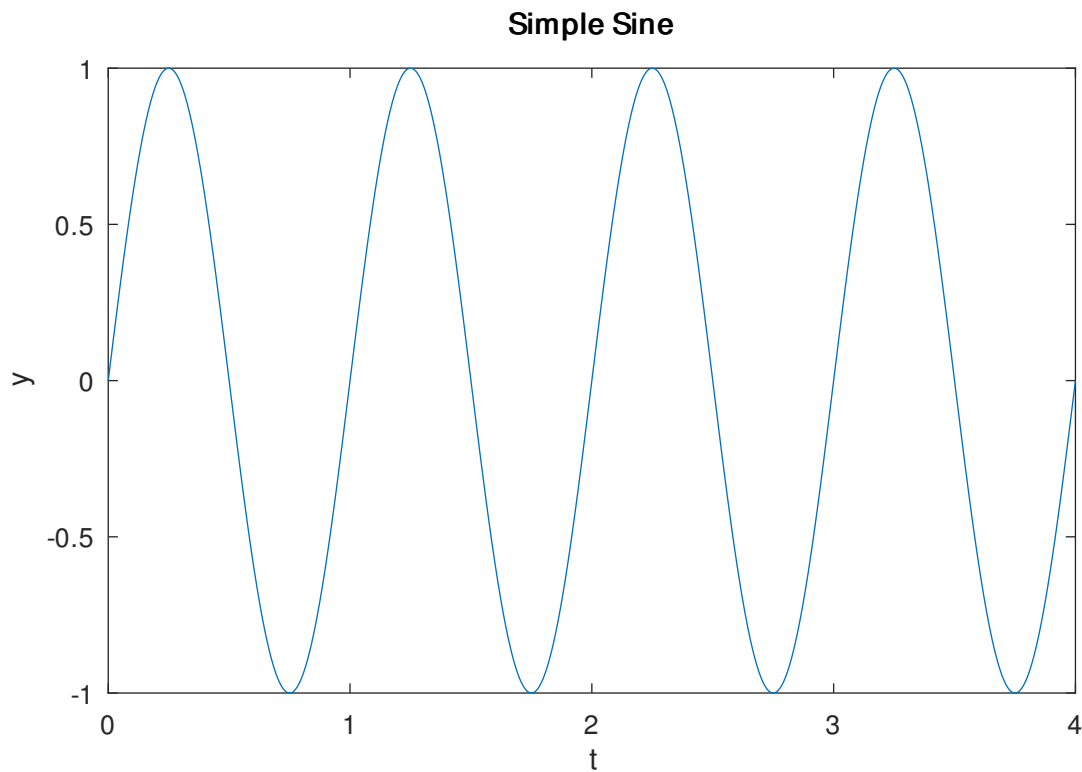


Figure 3: $y = \sin(2\pi t)$ over $0 \leq t \leq 4$, sampled at 1000 points.